

Experience

Independent Researcher & Open-Source Developer

Feb 2026 - Present

a9l.im

- Conduct independent interpretability and alignment research from hypothesis through experiments and writeup; publish open-source LLM tooling.

Selected Research

Saklas

Apr 2026 - Present

Mechanistic interpretability workbench - Python, PyTorch

- Built a local workbench for probing and steering LLMs along curved manifolds.
- Implemented Jacobian-lens readouts and arbitrary-role chat support.
- Published a typed Python API, dashboard, and OpenAI/Ollama-compatible server compatible with most open source LLMs. Available on PyPI with CUDA and MPS support.

Introspection via Kaomoji

Apr 2026 - Jun 2026

Emotional concept representation study - Python, Hugging Face

- Across five model families, the emotional category of a prompt was decodable with 99.2% accuracy from the hidden state; the opening kaomoji alone retained 80.6% accuracy on Gemma versus 11.1% by chance.
- Procrustes-aligned emotional category centroids to recover shared geometry; built kaomoji collection tooling and published the corpus on Hugging Face.

Elapsed-Time Encoding in LLMs

Jun 2026 - Present

Ongoing time encoding study - Python, PyTorch

- Showed elapsed conversational time is linearly encoded at an elicited token ($r = 0.88$, about 0.3 seconds per token) using linear probes.

Neuralese as a Conlang

Jul 2026 - Present

Ongoing abstract-CoT analysis study - Python, PyTorch

- Implemented Abstract-CoT to test if abstract tokens could be interpreted as a compositional language.
- A preliminary 135M run did not successfully learn to use abstract tokens to reason.

Block Sparse Crosscoders

Jul 2026 - Present

Ongoing cross-layer representation geometry study - Python, PyTorch

- Synthesized recent work on block sparse autoencoders and sparse crosscoders to extract cross-layer multidimensional features.

Selected Engineering

Geon

Feb 2026 - Present

Relativistic N-body simulator - JavaScript, WebGPU

- Built a WebGPU backend to render and simulate N-body post-Newtonian physics.
- Implemented Boris integration to approximate Einstein-Infeld-Hoffmann dynamics with scalar fields.

Ogdoad

Jun 2026 - Present

Research library for recreational mathematics - Rust, Python

- Built a library to explore quadratic forms over the surreal numbers and Conway's combinatorial games.
- Implemented Python bindings and a purpose-built language.

Other Projects

Feb 2026 - Present

Browser simulations and agentic tooling - JavaScript, WebGPU

- **Agent infrastructure.** Gaslamp, a cross-review CLI for Claude Code and Codex, and Dagwood, a version-controlled DAG-based task tracker with an MCP interface.
- **Additional interactive systems.** Built seven more browser projects across physics, finance, biology, political science, and comparative religion using vanilla JS.

Education

University of California, San Diego

Sep 2023 - Mar 2026

B.S. Mathematics - GPA 3.75 - completed early

Technical Skills

Research/ML: mechanistic interpretability; Python, PyTorch, NumPy/SciPy, scikit-learn, CUDA/MPS.**Engineering:** Java, JavaScript, WebGPU, Rust, FastAPI, Cloudflare Workers, MCP tooling, Git.